University of Glasgow

W1-02: Hardware Primer

COMPSCI4064 (H) and COMPSCI5088 (M)

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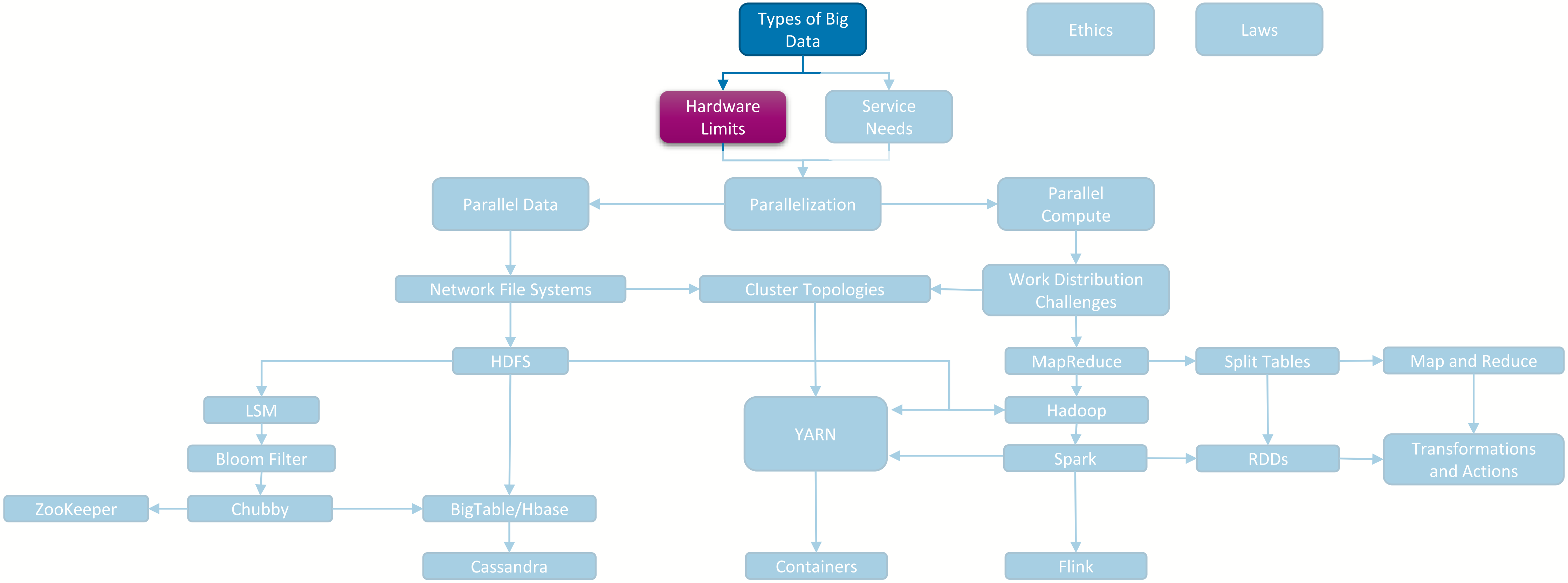
Where are we?



Types of Big Data
Ethics
Laws
Hardware Limits
Service Needs
Parallel Data
Parallelization
Parallel Compute
Network File Systems
Cluster Topologies
Work Distribution Challenges
HDFS
MapReduce
Split Tables
Map and Reduce
LSM
YARN
Hadoop
Bloom Filter
Spark
RDDs
Transformations and Actions
ZooKeeper
Chubby
BigTable/Hbase
Cassandra
Containers
Flink

Hardware Types

- One of the key factors that turn normal computing problems into Big Data problems is insufficient hardware
  - … or to put it another way, can I solve this problem on a laptop?

- Today there are four main hardware aspects we need to understand
  - **General Purpose Compute**: How much normal work the machine can do in a fixed period of time
  - **Specialist Compute**: How much hardware-accelerated work the machine can do
  - **Storage Stack**: How much of each type of storage the machine has
  - **Network**: How quickly can it get or send data

General Purpose Compute

- How quickly the CPU (Central Processing Unit) can complete a give task

- CPUs become faster as engineers manage to fit more transistors inside, as well as optimise the internal architecture of those transistors for common operations

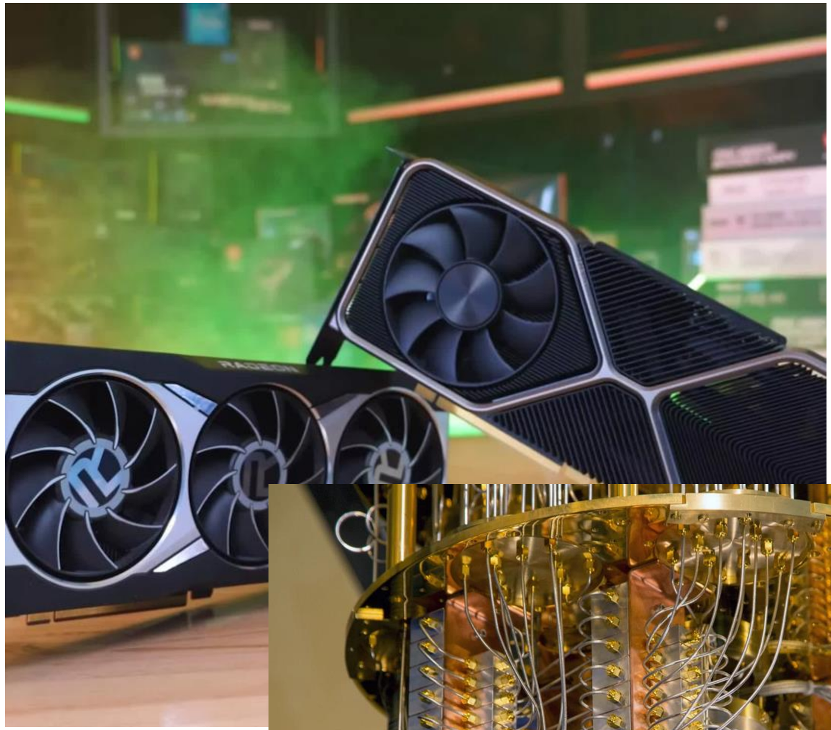
- Use representative benchmarks to evaluate compute capacity, not cores, clock speeds, or power consumption

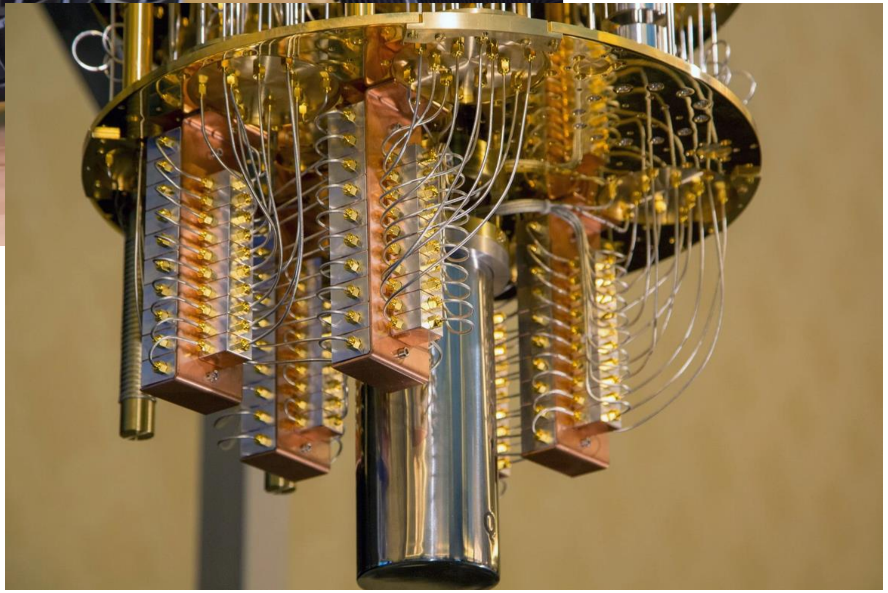
Specialist Compute

- Since the CPU is designed to work for a wide range of tasks, its often not optimised enough to complete particular workloads quickly, or it may not support workloads we care about

- For this reason we may use specialist hardware accelerators or add-ons to speed up compute

- Examples:
  - GPUs for Deep Learning
  - Quantum Cores

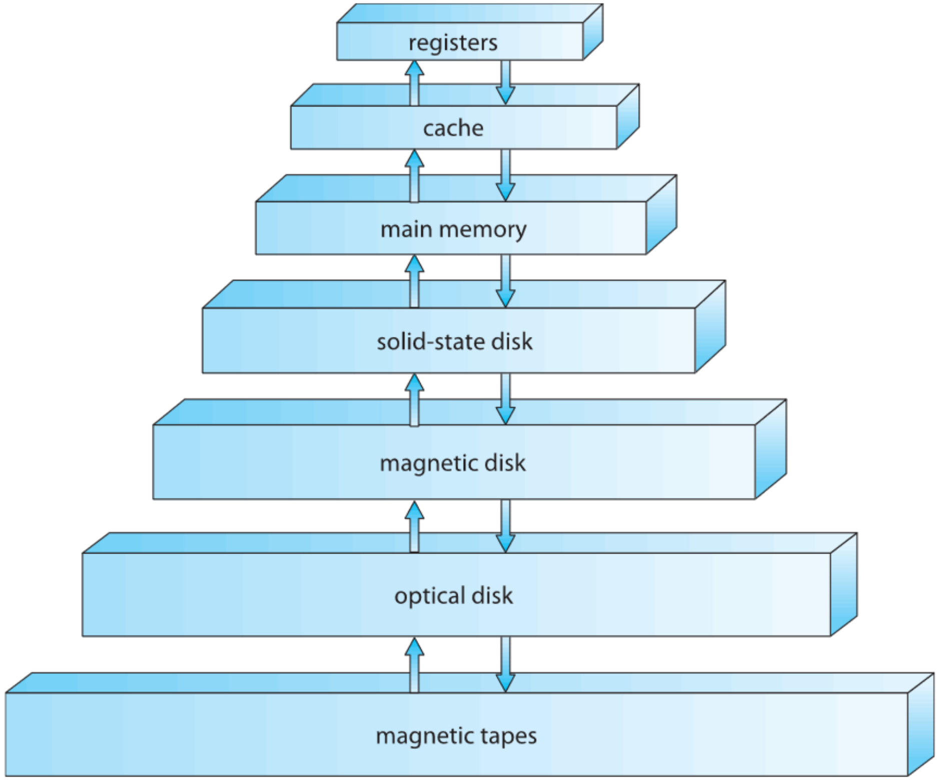




Storage Stack

- Inside a typical computer we have a wide range of types of tiered storage
  - Upper tiers are small and expensive, but fast
  - Lower tiers are large and cheap but slow

- **Registers**: Extremely fast, held within the CPU die, transient
- **Cache (L1-3)**: Very fast, directly connected to the CPU, transient
- **Main Memory (RAM)**: Fast, indirectly connected to the CPU via a memory controller, transient
- **Solid State Drives (SSD)**: Medium Speed, connected on the general data bus, but are persistent
- **Magnetic (Hard) Drives**: Slow, connected on the general data bus, and are persistent and cheap



Storage Stack

Source: A. Silberschatz, “Operating System Concepts”, 9th Ed., 2012.

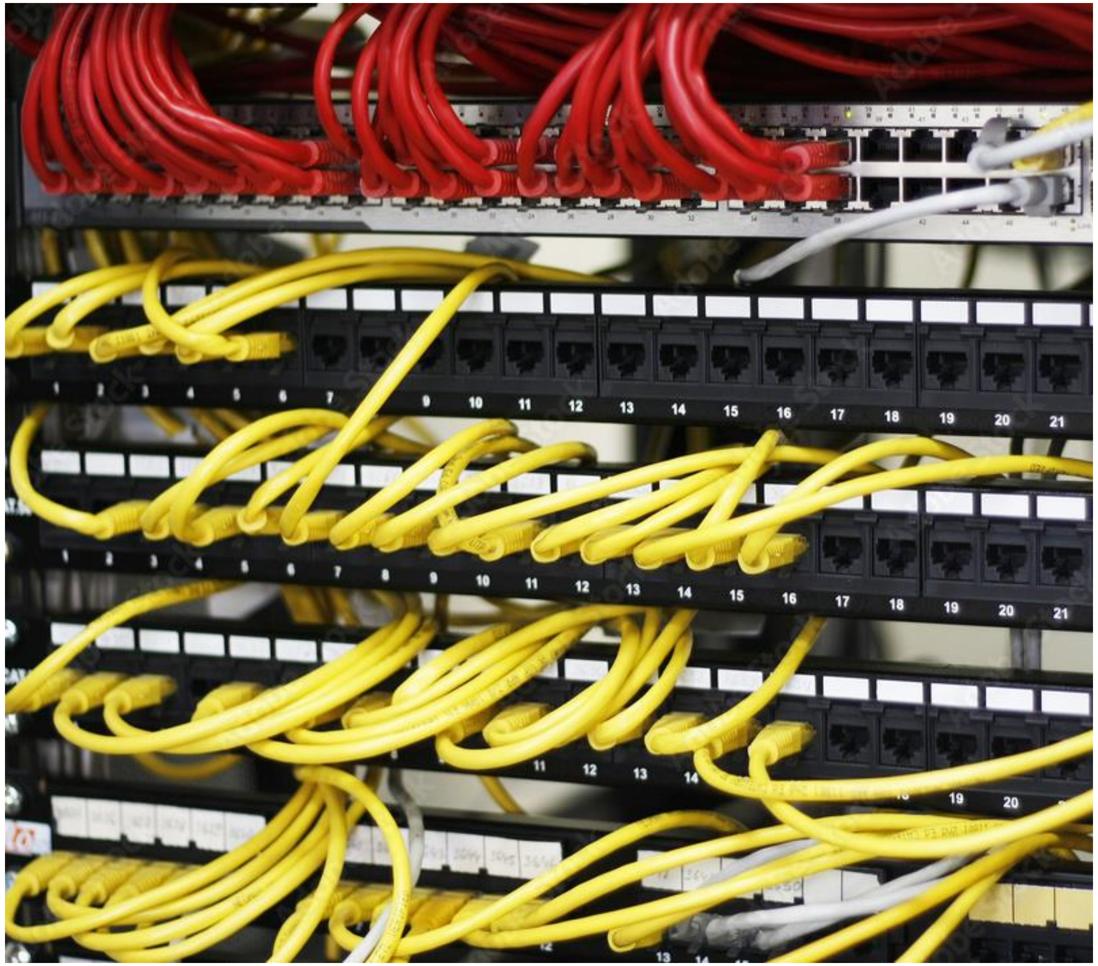
Storage Stack Speeds

Event	Latency	Scaled
1 CPU cycle	0.3 ns	1 s
Level 1 cache access	0.9 ns	3 s
Level 2 cache access	2.8 ns	9 s
Level 3 cache access	12.9 ns	43 s
Main memory access (DRAM, from CPU)	120 ns	6 min
Solid-state disk I/O (flash memory)	50–150 μs	2–6 days
Rotational disk I/O	1–10 ms	1–12 months

These layers have got a bit faster since this paper was written

B. Gregg, “Systems Performance: Enterprise and the Cloud”, Prentice-Hall, 2013.

Network



- Much of this course will be focused on distribution of work across multiple machines
- Hence these machines will need to communicate – incurring transfer latencies across the local area network (LAN)
- Individual machines are physically connected to central routers/switches

Network

- Hence, we also need to care about the costs of sending and receiving data across a network

- Network transfers are nearly always slower than local storage
  - (there are exceptions in specialised server farms)

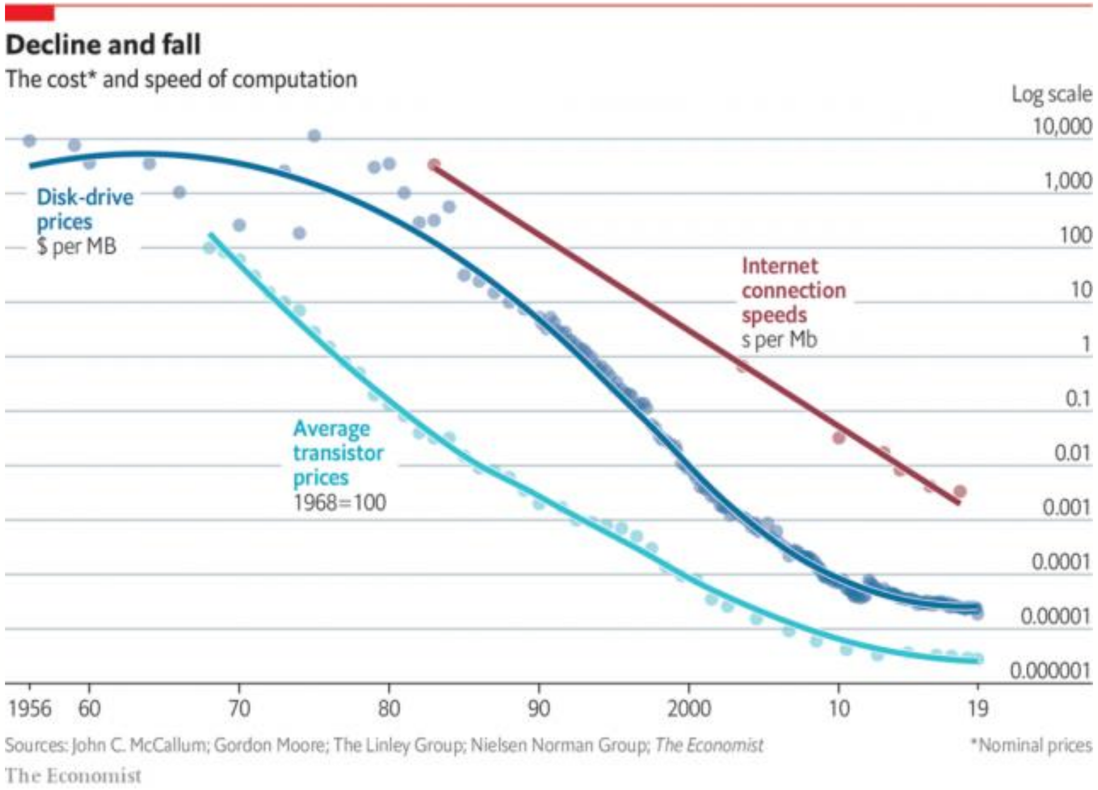
Event	Latency	Scaled
LAN: Transfer Latency	1 ms	1 years
Internet: San Francisco to New York	40 ms	4 years
Internet: San Francisco to United Kingdom	81 ms	8 years
Internet: San Francisco to Australia	183 ms	19 years
TCP packet retransmit	1–3 s	105–317 years
OS virtualization system reboot	4 s	423 years
SCSI command time-out	30 s	3 millennia
Hardware (HW) virtualization system reboot	40 s	4 millennia
Physical system reboot	5 m	32 millennia

B. Gregg, “Systems Performance: Enterprise and the Cloud”, Prentice-Hall, 2013.

Cost

- In general, the cost of hardware has been decreasing over time

- However, the pace of this cost reduction has slowed dramatically for compute and storage



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